

AERODYNAMICS

Preliminary aerodynamics design mainly figure out the value of aerodynamic characteristics (i.e. Wing Span, Wing Surface Area, Aspect Ratio, C_L , C_D , C_m , and Aerodynamic Center), and takeoff, turning, and landing performance.

Assumptions

- Performing at elevation of 1300ft where Wichita at
- The data on table1 is trustable
- Load factor to be 4

Aerodynamic Characteristics

Wing span: The wing span is a key parameter for aspect ratio (AR). Higher the wing span increases the aspect ratio. We desired to have 4ft in the conceptual design.

Wing Surface Area: The wing surface area is the key parameter for the stall velocity, wing loading, and total lift. Increasing wing surface area increases total lift and decreases wing loading (W/S), which reduces stall velocity. This enables a lightweight propulsion system due to low thrust-to-weight (T/W) requirements. We desired to have around 4ft^2 in the conceptual design.

Aspect Ratio: The aspect ratio affects the $C_{L\alpha}$ and $C_{D\alpha}$. According to Kuchemann's modified Helmbold's equation^A which is applied to convert C_i to C_L , and finite wing theory^B which is applied to get C_d to C_D , higher aspect ratio increases $C_{L\alpha}$ and possibly decreases $C_{D\alpha}$. Our desired value was 5 in the conceptual design.

Aerodynamic Center: Aerodynamic center is calculated by using $C_{L\alpha}$ and $C_{m\alpha}$. This point is considered where lift, drag, and moment generate on. So, this point will be important for structure and S/C. This value is usually about 25% of chord length.

Detailed aerodynamic characteristics are shown on table1. Aspect ratio is lower than what we desired, but this value is based on actual wing span and wing surface area. So, we expect reference aspect ratio to be higher than this; we have complex wing shape.

Use of XFOIL^C

Description

XFOIL code is an interactive program which utilizes a two-dimensional panel method code with integral boundary layer theory to analyze airfoil in viscous or inviscid flow fields. The XFOIL code also incorporates a Karman-Tsien compressibility correction, which gives accurate flow prediction until sonic condition. XFOIL has proven accurate in the low Reynolds Number regime, but no less than 100000, and can predict separation bubbles. The code includes a laminar/turbulent transition criterion using special amplification theory. Moreover, the aerodynamic data given by XFOIL does not consider a unit chord length. So, to get comparable data with experimental data, an airfoil geometry loaded in to XFOIL must be 1.

The XFOIL code is one of easy way to predict C_l and C_m value, especially at the linear region (Figure1). In fact, XFOIL C_l line is a little bit higher than its experimental data^D, but linear slope of C_l line based on XFOIL data below zero angle of attack is perfectly matched. For some unknown reason, there is a little C_l value decrease around zero angle of attack on experimental data, but there should not be any decrease around this area because the value should linearly increased around this area. So, as long as XFOIL's linearized slope matched with experimental data below zero angle of attack, rest of the linear region should trustable. On the drag polar (Figure2), XFOIL data is higher than its experimental value. This implies that XFOIL's C_d data is no trustable, but overestimation is better than underestimation in the

case of drag. So, even though the value is not trustable, those values can be used to estimate drag. Overall, XFOIL data can be used for estimating coefficients without having any test.

Takeoff and Landing Performance

Takeoff Distance: Key parameters of takeoff distance are $C_{L\ MAX}$, thrust, wing surface area, aircraft weight, and Lift and Drag at 0.7 times of takeoff velocity which is 1.2 times of stall velocity^B. At this point, we know the $C_{L\ MAX}$, wing surface area, aircraft weight, and Lift and Drag (Table1), so thrust is the parameter that can change the takeoff distance (Figure3).

Landing Distance: Key parameters of landing distance are $C_{L\ MAX}$, wing surface area, aircraft weight, and Lift and Drag at 0.7 times of landing velocity which is 1.3 times of stall velocity^B. We know all the parameters at this point, thrust is not a key parameter for landing. So, we can estimate the landing distance (Table1).

Turning Performance

Turn Radius: Key parameters of turn radius are $C_{L\ MAX}$, aircraft weight, wing surface area, and load factor^B. As long as aircraft weight is the parameter, turn radius will be changed before and after the mission (Table1).

Bank Angle: Bank angle is calculated by load factor^B. So, the angle will not change before and after the mission.

Elevation (ft)	1300
Density (slugs/ft ³)	2.288×10^{-3}
Viscosity (slugs/ft*s)	3.71×10^{-7}
Airfoil	NACA4418
Reynolds Number	250,000
Takeoff Weight (lb)	7.5
Landing Weight (lb)	5.5
Wing Span (ft)	4.5
Wing Surface Area (ft ²)	5.17
Aspect Ratio	3.91
Aerodynamic Center (%)	24.47
$C_{L \text{ MAX}}$	1.006
$C_{L \alpha=0}$	0.278
$\alpha_{CL=0}$ (degree)	-4.313
$C_{D,0}$	0.019
$C_{m,0}$	-0.096
V_{cruise} (fps)	69.71
V_{stall} (fps)	36.64
V_{takeoff} (fps)	43.97
V_{landing} (fps)	47.63
Landing Distance (ft)	85.33
Turn Radius before mission	40.71
Turn Radius after mission (ft)	29.85
Bank Angle (degree)	75.52

Table1, Data List

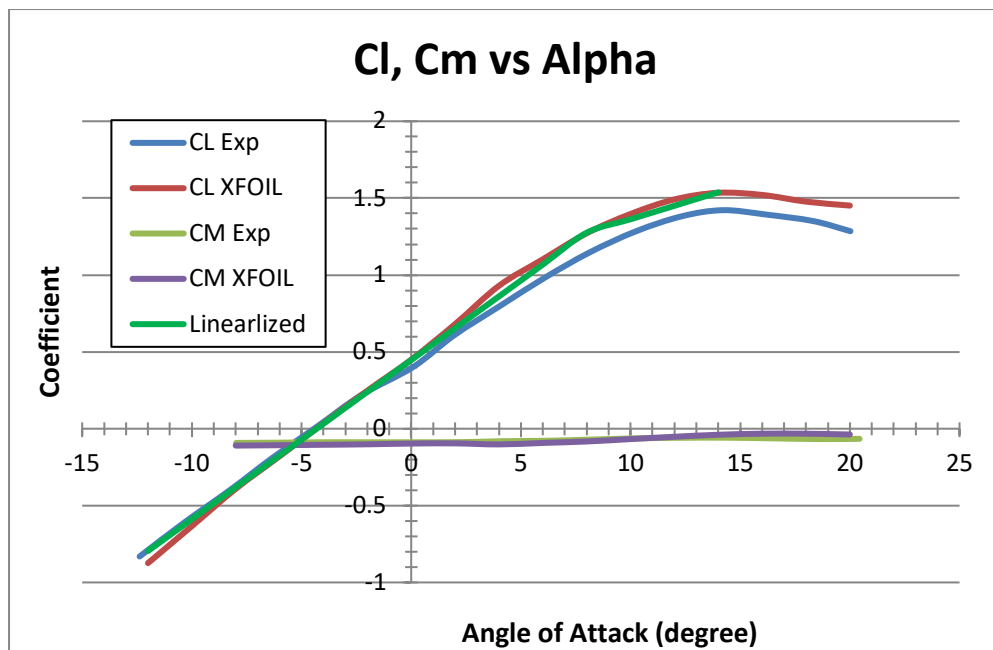


Figure1, C_l and C_m Comparison with Experimental Data and XFOIL Data

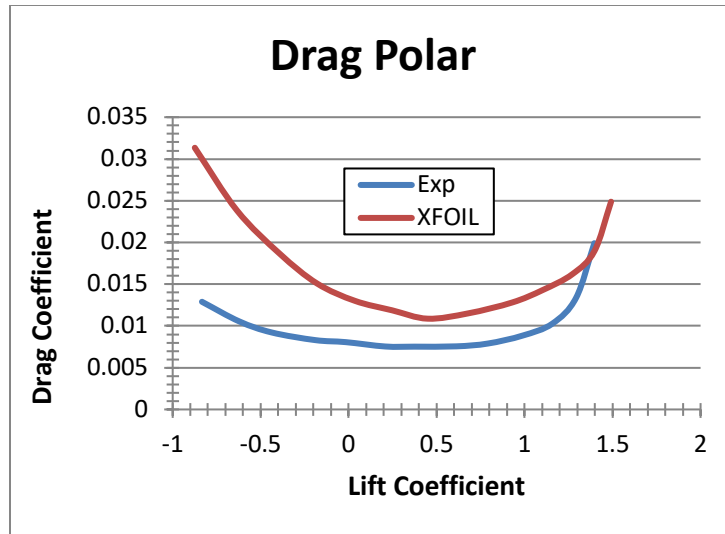


Figure2, Drag Polar Comparison with Experimental Data and XFOIL Data

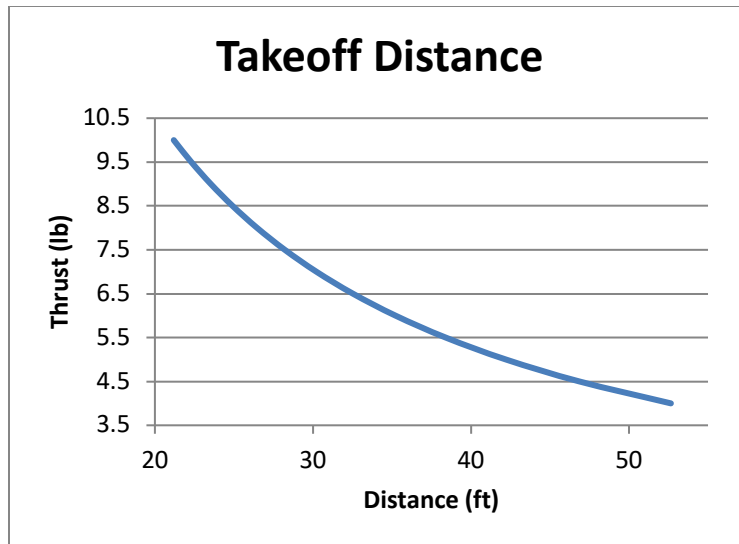


Figure3, Takeoff Distance Difference by having Different Thrust

Reference

- A. Anderson, John. *Introduction to Flight*. 5th Edition. Boston: McGraw Hill, 2005. Print.
- B. Anderson, John. *Fundamentals of Aerodynamics*. 4th Edition. Boston: McGraw Hill, 2005. Print.
- C. Drela, M., XFOIL. Vers. 6.94. Computer software. MIT, 2001. Windows XP, 532 KB, Download.
- D. Ira H. Abbott, Albert E. von Doehoff, and Louis S. Silvers, Jr., *NACA Report No. 824 Summary of Airfoil Data*. p.277-288

Nomenclature

α	Angle of attack
$\alpha_{CL=0}$	Angle that lift coefficient is zero
C_l	Airfoil lift coefficient
$C_{l\alpha}$	Airfoil lift curve slope
C_m	Airfoil pitching moment coefficient
C_{m0}	Airfoil pitching moment coefficient at zero angle of attack
$C_{m\alpha}$	Airfoil pitching moment curve slope
$C_{L\max}$	Wing maximum lift coefficient
$C_{L\alpha=0}$	Wing Lift coefficient at zero angle of attack
$C_{D,0}$	Drag coefficient at zero angle of attack for wing
W/S	Wing loading
T/W	Thrust-to-weight ratio
AR	Aspect ratio
V_{stall}	Stall velocity
V_{cruise}	Cruise velocity
V_{takeoff}	Takeoff velocity
V_{landing}	Landing velocity